Independence and discrete random variables

Events · Variables · Counts

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Probability and Statistics for Engineers

October 5, 7, and 9, 2026

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

HW 1	**Tue, Oct 6**	Gradescope
Quiz 2	**Sun, Oct 11**	Canvas
HW 2	**Mon, Oct 12**	Gradescope

Midterm 1: Fri, Oct 16, during class.

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



Monday: independence and random variables

From Week 1

$$P(A \cap B) = P(A \mid B)\, P(B), \quad P(B) > 0.$$

Conditioning can change a probability.

Today

1. Decide when it does not change.
2. Assign numbers to outcomes.
3. Translate numerical statements into events.

Check the conditioning

Draw two cards without replacement from a standard deck. If the first is an ace, what is the chance the second is an ace?

Independent events

Definition

Events A and B are independent if

$$P(A \cap B) = P(A)\,P(B).$$

If $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$.

Derive the equivalence from the conditional-probability formula.

The product definition still applies when one of the probabilities is zero.

Disjointness and independence are different

Disjoint: $A \cap B = \varnothing$. Independent: $P(A \cap B) = P(A)\,P(B)$.

For two independent fair tosses, $\Omega = \{HH, HT, TH, TT\}$.

Events	$P(A \cap B)$	$P(A)\,P(B)$
$A = \{HT\},\ B = \{TH\}$	0	1/16
$A =$ first heads, $B =$ second heads	1/4	1/4

Which pair is independent? Which pair is disjoint?

Disjoint events with positive probabilities cannot be independent. If one event has probability zero, the product and intersection probability are both zero.

A random variable assigns a number to each outcome

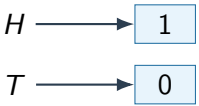
Definition

A random variable is a real-valued function on the sample space:

$$X : \Omega \longrightarrow \mathbb{R}, \qquad \omega \longmapsto X(\omega).$$

Example: one coin toss

$$\Omega = \{H, T\}.$$



The rule and its value

$$X(H) = 1, \qquad X(T) = 0.$$

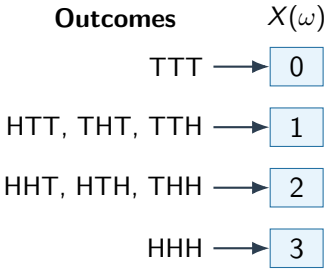
Once the outcome ω is known, the number $X(\omega)$ is determined.

$$\{X = 1\} = \{H\}, \qquad P(X = 1) = P(\{H\}).$$

For every real t, $\{\omega \in \Omega : X(\omega) \leq t\}$ must be an event, so its probability is defined. This holds automatically on a finite sample space when every subset is an event.

A value of a random variable describes an event

Three independent fair coin tosses. Let X count the heads.



The notation $\{X = 2\}$ means

$$\{\omega \in \Omega : X(\omega) = 2\}.$$

Here this event is

$$\{\mathrm{HHT}, \mathrm{HTH}, \mathrm{THH}\}.$$

Each sequence has probability $1/8$.

Several outcomes can give the same value. Add the probabilities of those outcomes.

An indicator records whether an event occurs

For an event $A \subseteq \Omega$, define

Indicator of A

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

$$\{I_A = 1\} = A, \qquad \{I_A = 0\} = A^c.$$

One fair die. Let $A = \{2, 4, 6\}$. List the values of I_A and find $P(I_A = 1)$.

The random variable records a fact about an outcome; it does not change the experiment.

Practice: translate a numerical condition into an event

Roll one fair die. Let $X(\omega)$ be its face value and set $Y(\omega) = I_{\{2,4,6\}}(\omega)$.

1. Describe $\{X \leq 3\}$ and $\{Y = 1\}$ as sets of outcomes.
2. Find $P(X \leq 3)$, $P(Y = 1)$, and $P(X \leq 3, Y = 1)$.
3. Are the two events independent?

Begin with the outcomes, then compute their probabilities.

Events, variables, and values

Object	Meaning
$\omega \in \Omega$	An outcome of the experiment.
$X : \Omega \to \mathbb{R}$	A fixed rule assigning a number to each outcome.
$X(\omega)$	The value obtained at that outcome.
$\{X \leq t\}$	The event $\{\omega : X(\omega) \leq t\}$.
I_A	A variable recording whether A occurs.

Independence is a probability condition; it is not the same as disjointness.

Wednesday: from an outcome to a distribution

Recall the three-toss count

$$X = \text{number of heads.}$$

$$\Omega = \{HHH, HHT, \ldots, TTT\}.$$

Today

1. Collect probabilities into a PMF.
2. Compare the PMF with its CDF.
3. Read probabilities with the correct endpoints.

Two different objects

HHT is an experimental outcome. The number 2 is the value of X at that outcome.

Check: name a different outcome with the same value of X.

Probability mass belongs to values

A discrete random variable takes values in a finite or countable set.

Probability mass function (PMF)

$$p_X(x) = P(X = x).$$

For its positive-mass values $S_X = \{x : p_X(x) > 0\}$,

$$p_X(x) \geq 0, \qquad \sum_{x \in S_X} p_X(x) = 1.$$

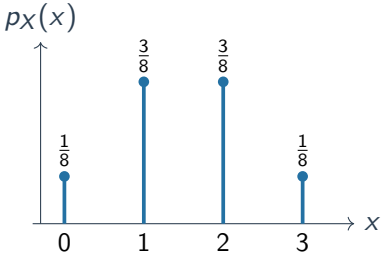
Why must the sum be one? Use the events $\{X = x\}$.

$I_{\{C\}}$ is 1 when condition C holds and 0 otherwise. For a fair die, $p_X(x) = \frac{1}{6} I_{\{x \in \{1,\ldots,6\}\}}$: only the six faces receive mass.

The same distribution as a table and a PMF plot

For the number of heads in three independent fair tosses:

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8



Read probabilities from masses.

$$P(X \in \{1, 2\}) = p_X(1) + p_X(2).$$

The count is integer-valued: $P(X = 1.5) = 0$.

The CDF accumulates probability up to a threshold

Cumulative distribution function (CDF)

For every real t,

$$F_X(t) = P(X \leq t).$$

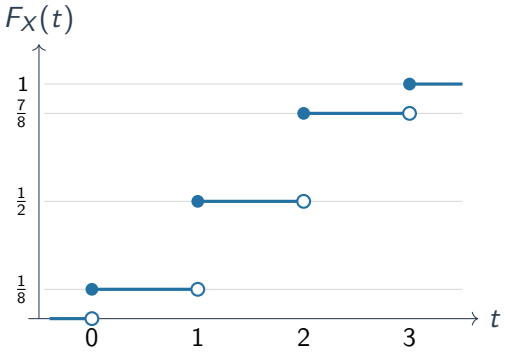
For a discrete X, this is $F_X(t) = \sum_{x \leq t} p_X(x)$.

For the three-toss count, fill in the cumulative values.

The threshold t may be noninteger even when X is integer-valued.

A discrete CDF is a step function

$$F_X(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{8}, & 0 \le t < 1, \\ \frac{1}{2}, & 1 \le t < 2, \\ \frac{7}{8}, & 2 \le t < 3, \\ 1, & t \ge 3. \end{cases}$$



Filled dots give the function value. The jump at k is $P(X = k)$.

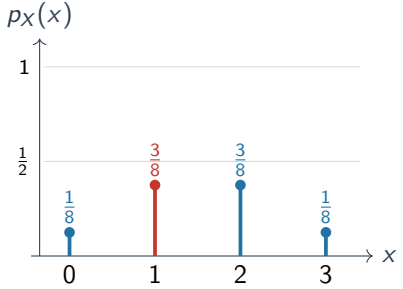
A CDF is nondecreasing and right-continuous, with limits 0 and 1 at $-\infty$ and $+\infty$.

PMF and CDF of the same random variable

X: number of heads in three independent fair tosses.

PMF: mass at a value

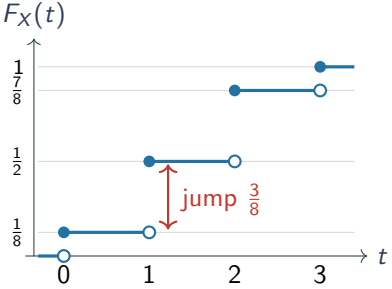
$$p_X(x) = P(X = x)$$



$$p_X(1) = \tfrac{3}{8}.$$

CDF: mass up to a threshold

$$F_X(t) = P(X \leq t)$$



$$F_X(1) = \tfrac{1}{8} + \tfrac{3}{8} = \tfrac{1}{2}.$$

At 1, the PMF gives the CDF's jump: $\frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. Between possible values, the CDF stays constant.

Read the endpoints before subtracting CDF values

For real $a < b$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Split $\{X \leq b\}$ into $\{X \leq a\}$ and $\{a < X \leq b\}$.

For the three-toss count, compare:

$$P(X < 1), \qquad P(X \leq 1), \qquad P(1 < X \leq 2).$$

- P(X<1)=P(X<=0)=F(0)
- P(X<=1)=F(1)
- P(1<X<=2) = F(2)-F(1)

For integer-valued X and integer k: $P(X < k) = F_X(k-1)$; $P(X > k) = 1 - F_X(k)$.

Practice: recover a CDF from three masses

x	-1	1	4
$p_X(x)$	$1/4$	$1/2$	$1/4$

1. Sketch $F_X(t)$; mark its value at each jump.
2. Find $P(X = 1)$, $P(X \leq 1)$, and $P(-1 < X \leq 4)$.

Point mass and accumulated mass

PMF

$$p_X(x) = P(X = x).$$

One value; nonnegative masses adding to one.

CDF

$$F_X(t) = P(X \le t).$$

All values at or below a real threshold.

$$P(a < X \le b) = F_X(b) - F_X(a), \qquad a < b.$$

For our integer-valued examples, the jump at k is $p_X(k)$.

A CDF is nondecreasing and right-continuous. Its limits are 0 and 1 at $-\infty$ and $+\infty$.

Friday: a model for repeated trials

From the previous lessons

- Independence permits probabilities to factor.
- A random variable assigns a value to each outcome.
- A PMF assigns probabilities to those values.

Today

Count successes in a fixed number of independent trials.

$$\text{one pattern's probability} \quad \times \quad \text{number of patterns.}$$

We will derive the binomial formula using independence and counting.

One Bernoulli variable and a count

For an event A_i with $P(A_i) = p$, define its indicator I_i.

Trial i

$$I_i = \begin{cases} 1, & \text{success}, \\ 0, & \text{failure}. \end{cases}$$

$$p_{I_i}(u) = (1-p)I_{\{u=0\}} + pI_{\{u=1\}}.$$

This is the Bernoulli law: $I_i \sim \mathrm{Bern}(p)$.

Count in n trials

$$X = I_1 + \cdots + I_n.$$

$$X \in \{0, 1, \ldots, n\}.$$

The event $\{I_i = 1\}$ is A_i. If the success events are mutually independent with common probability p, these are Bernoulli trials.

For the derivation, take $n \geq 1$ and $0 < p < 1$. The cases $p = 0$ and $p = 1$ give constant counts.

Independent trials: what the assumption means

Bernoulli trials

A fixed number $n \geq 1$ of trials, each with success probability p. The trials are mutually independent: the probability of every prescribed set of trial results factors into the corresponding individual probabilities.

$$P(I_1 = i_1, \ldots, I_n = i_n) = \prod_{j=1}^{n} P(I_j = i_j), \qquad i_j \in \{0, 1\}.$$

Check the full assumption

Pairwise independence alone does not justify this factorization.

For $0 < p < 1$, observing earlier results does not change a later trial's success probability.

First calculate the probability of one pattern

Four independent trials, each with success probability p.



Specified pattern: success, failure, success, failure.

For any specified pattern with k successes in n trials

$$\text{pattern probability} = p^k(1-p)^{n-k}.$$

Then count the patterns

Let X count successes in n Bernoulli trials.

1. Choose the k positions occupied by successes.
2. Every such pattern has probability $p^k(1-p)^{n-k}$.
3. Distinct patterns are disjoint outcomes, so add their probabilities.

Binomial distribution: $X \sim \mathrm{Bin}(n, p)$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} I_{\{k \in \{0,\ldots,n\}\}}.$$

Evaluate the formula only for $k = 0, \ldots, n$; set it to zero elsewhere.

The indicator includes the integer restriction. It never asks us to evaluate an invalid factorial.

Exactly two successes or at least one?

A hypothetical randomized test

Four independent runs each reveal a particular defect with probability $1/4$. Let X count the runs that reveal it. Thus $X \sim \mathrm{Bin}(4, 1/4)$.

Exactly two runs

At least one run

The complement is a single pattern: four failures.

Check the assumptions before choosing a binomial model

A binomial count has

a fixed number of trials, two outcomes per trial, a common p, and mutual independence.

Sampling without replacement

A bag has three red balls and two blue balls. Draw two balls without replacement. Let X count red balls. Is $X \sim \mathrm{Bin}(2, 3/5)$?

A fixed trial count and two outcomes are not enough.

Practice: count heads in three fair tosses

Let X be the number of heads in three independent fair coin tosses.

1. Find $P(X = k)$ for $k = 0, 1, 2, 3$.
2. Explain the symmetry between k and $3 - k$.

This recovers Wednesday's table from a general formula.

Independence and counts: the essential distinctions

Question	What to check
Independent?	$P(A \cap B) = P(A)\,P(B)$.
Disjoint?	$A \cap B = \varnothing$; a condition on events, not factorization.
Several trials?	Pairwise independence alone does not justify a full pattern product.
Binomial count?	Fixed n, common p, and mutually independent Bernoulli trials.
At least one?	Consider $1 - P(\text{none})$.

State the model before substituting into a formula.

Optional: taking a complement preserves independence

Assume A and B are independent.

Claim

A and B^c are also independent.

Split A into the disjoint pieces $A \cap B$ and $A \cap B^c$.

Repeating this argument also gives independence of A^c and B^c.

Optional: pairwise independence does not imply mutual independence

Two independent fair coins. A: first H; B: second H; C: results agree.

Outcome	A	B	C
HH	1	1	1
HT	1	0	0
TH	0	1	0
TT	0	0	1

Each row has probability $1/4$.

Compare pairs with the triple.

Mutual independence of $A_1, \ldots, A_n$

For *every* subset $J \subseteq \{1, \ldots, n\}$ with $|J| \geq 2$,

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j).$$

References for this week

DeGroot and Schervish

Probability and Statistics, fourth edition.

- §2.2: independence.
- §§3.1, 3.3: discrete random variables and CDFs.
- §5.2: the binomial distribution.

Mor Harchol-Balter

Introduction to Probability for Computing, supplied 2024 version.

- Chapters 2–3: probability rules and discrete random variables.

Sheng Jiang

STAT 131 Notes, July 18, 2024: pp. 21–26 and 30–31.

Figures and exposition retain Antonio de Leon's earlier STAT 131 teaching style.